WEEK 7

NLP in Pharma Commercial
Adverse events, field notes and social listening

Detecting adverse events in free text, explaining the model to a safety reviewer, and reusing the same tools on the rest of what the field writes down.

Andi Cupallari · Applied Data Science and AI in Pharma Commercial · Fall 2026 · Monday, October 26

AGENDA

Tonight

1	**Text and the rules**	Where the text comes from; what counts as an adverse event; when the clock starts
2	**The toolbox**	Bag of words, embeddings, transformers; precision and recall
3	**Building the detector**	Lexicon, then logistic regression; choosing the threshold
4	**Explaining the model**	Coefficients, word contributions, LIME, SHAP, leak checks
5	**From flag to case**	Four elements, seriousness, due date, PHI
6	**Other uses of the same tools**	Topics, social listening, MI routing, signal detection
7	**Close**	GenAI in one slide; governance; exercise

01

Text and the rules

What gets written down, and what the regulator expects us to do with it

Three streams of text reach a brand team

Field call notes

- Typed by the rep after the visit
- Short, abbreviated, inconsistent
- 4,200 notes · 60 reps · 2,400 HCPs
- Company knowledge the moment they are saved

Patient forum posts

- A company-sponsored community and an independent forum
- First person, lay language
- 3,000 posts
- Duty to screen depends on who runs the site

Medical-information inquiries

- Phone, email, web form
- HCPs, pharmacists, patients, caregivers
- 1,200 inquiries
- Routed by intent; safety intake on all

~80%
of EMR data is unstructured

34M
clinical notes mined in one NLP chart-review program

87%
fewer papers screened by hand with AI-assisted review

94%
median share of adverse drug reactions never reported

Source: IQVIA, Harnessing the Power of AI in Medical Affairs (2025); Hazell & Shakir, Drug Safety 2006.

1 · ONE NOTE

One call note, four jobs

Dr. reports BCBS is requiring prior authorization for GLUCOREN; two patients waiting over a week. Left PA form and hub phone number. Side note: a 58-year-old female patient reported dizziness within two weeks of starting; HCP managing conservatively. HCP also asked whether GLUCOREN could be used in a type 1 patient — referred to Medical Information. Mrs. Kowalski to call the office back.

Access insight
PA barrier at BCBS → payer team

Adverse event
patient + event + drug + reporter → safety intake; clock running

Off-label request
unsolicited → Medical Information; rep does not engage

PHI
a patient name in free text → scrub before storage

Tonight builds the second row. The others use the same methods with different labels.

Source: Synthetic note, composed for illustration.

What the regulation says

Term	21 CFR 314.80	For the model
Adverse drug experience	Any adverse event associated with the use of a drug in humans, whether or not considered drug related	Association opens a case; causality is not the model's job
Serious	Death; life-threatening; hospitalization; persistent disability; congenital anomaly; other medically important event	A second label
Unexpected	Not listed in the current labeling	Lookup against the label
15-day Alert report	Serious and unexpected: report within 15 calendar days of initial receipt	The deadline the queue computes
Periodic report	Everything else	Still a case

> *FDA "will consider an applicant responsible for information known to its employees, affiliates, and contractors." A saved call note is company knowledge.*
>
> — FDA draft guidance, Postmarketing Safety Reporting (2001), §X

Source: 21 CFR §314.80; ICH E2D (2003).

A case needs four things

Identifiable patient
"a 58-year-old female", "one of his patients"

→

Identifiable reporter
the HCP, a caller, a forum member

→

Suspect drug
GLUCOREN, implicit in a brand call

→

Adverse experience
the event, in any words

Not a case

- **"No reports of nausea"** (negation)
- **"Asked about the nausea rate in the label"** (no patient)
- **"Worried about side effects before starting"** (hypothetical)
- **A competitor's patient** (not our product)

So the pipeline answers, in order

- **AE mentioned?** classifier
- **Four elements?** rules
- **Serious? Unexpected?** lexicon and label lookup
- **Due date** awareness date + 15 days if serious and unexpected

Source: FDA draft guidance (2001), §IV.B; ICH E2D.

Which text you are obliged to look at

FDA (2001 draft guidance, §VI.G)

- "Applicants should review any Internet sites sponsored by them for adverse experience information." Sites they do not sponsor: no duty to review; a duty to act on what they become aware of.

EMA GVP Module VI (Rev 2, VI.B.1.1.4)

- MAHs "should regularly screen internet or digital media under their management or responsibility." Media outside their control: act on awareness.

Stream	Who runs it	Duty
Aspen "Living with T2D" community	the company	screen regularly
T2D Circle forum	independent	act on what you see
Field notes, MI inquiries	the company	company knowledge

Running social listening on a site you do not control is a decision to acquire knowledge you must then act on. Route cases to Safety before anything goes to a dashboard.

Source: FDA (2001) §VI.G; EMA GVP Module VI Rev 2 (2017).

02

The toolbox

Text into numbers, and how to judge the result

Three ways to represent a sentence

	Bag of words / TF-IDF	Static embeddings	Contextual transformers
Idea	count terms; weight rare ones up	one vector per word, learned from co-occurrence	one vector per token in context, pretrained on large corpora
Knows that	“nausea” ≠ “vomiting”	“nausea” ≈ “vomiting”	“no nausea” ≠ “nausea”
Needs	labels	a corpus or a download	a download, a GPU to fine-tune, labels
Explained by	coefficients	nearest neighbours	Integrated Gradients; attention with care

Model	Pretrained on	Use for
BERT (Devlin 2019)	Wikipedia, books	general English: rep notes, patient posts
BioBERT (Lee 2020)	PubMed, PMC	literature, biomedical NER
ClinicalBERT (Alsentzer 2019)	MIMIC clinical notes	chart review
Sentence-BERT (Reimers 2019)	sentence pairs	search, clustering, BERTopic

Source: Pang, Lee & Vaithyanathan 2002; Deerwester et al. 1990; Devlin et al. 2019; Lee et al. 2020; Alsentzer et al. 2019; Reimers & Gurevych 2019.

Four rungs

Rules / lexicon
minutes to build · no labels · transparent

→

Classical ML
TF-IDF + logistic regression · labels · exact explanations

→

Fine-tuned transformer
BioBERT head · labels + GPU · best on hard text

→

LLM, zero-shot
a prompt · no labels · cost per call · still measured

Climb when

- error analysis shows negation, paraphrase or abbreviations the rung below cannot handle;
- you have labels;
- the gain holds on a held-out set.

Stay when

- the text is short and formulaic (CRM notes usually are);
- the reviewer needs to see why;
- the model has to be documented in a validated process.

Tonight stops at rung two and shows how to read rungs three and four when a vendor brings them.

Source: Karimi et al., ACM Computing Surveys 2015; Rudin, Nature Machine Intelligence 2019.

Judge a detector by its operating point

Precision

- Of the notes we flag, how many are real. Low precision wastes reviewer time.

Recall

- Of the real AEs, how many we catch. Low recall means unreported cases. In safety this comes first.

The curve

- Each threshold is a (recall, precision) pair. AUC and AP compare models; the chosen point is the decision.

Write down: target recall, resulting precision, flags per week, reviewer hours, cases missed per year. F1 hides all of that.

03

Building the detector

A lexicon to find the failure modes, then a model that reads context

Lexicon of 77 lay terms: recall 79%, precision 42%

Method

▸ Flag a note if any lay term appears (nausea, threw up, sugar lows, hospitalized…). Score against the PV reviewer's labels on the reviewed notes.

What a lexicon cannot tell

▸ **Who:** a patient, or the label?
▸ **Whether:** present, denied, hypothetical?
▸ **Spelling:** typos, abbreviations, paraphrase

False positives: word present, no event

▸ “no reports of yeast infection”
▸ “asked how common hypoglycemia was in the trials”
▸ “patient education materials on hypoglycemia awareness”

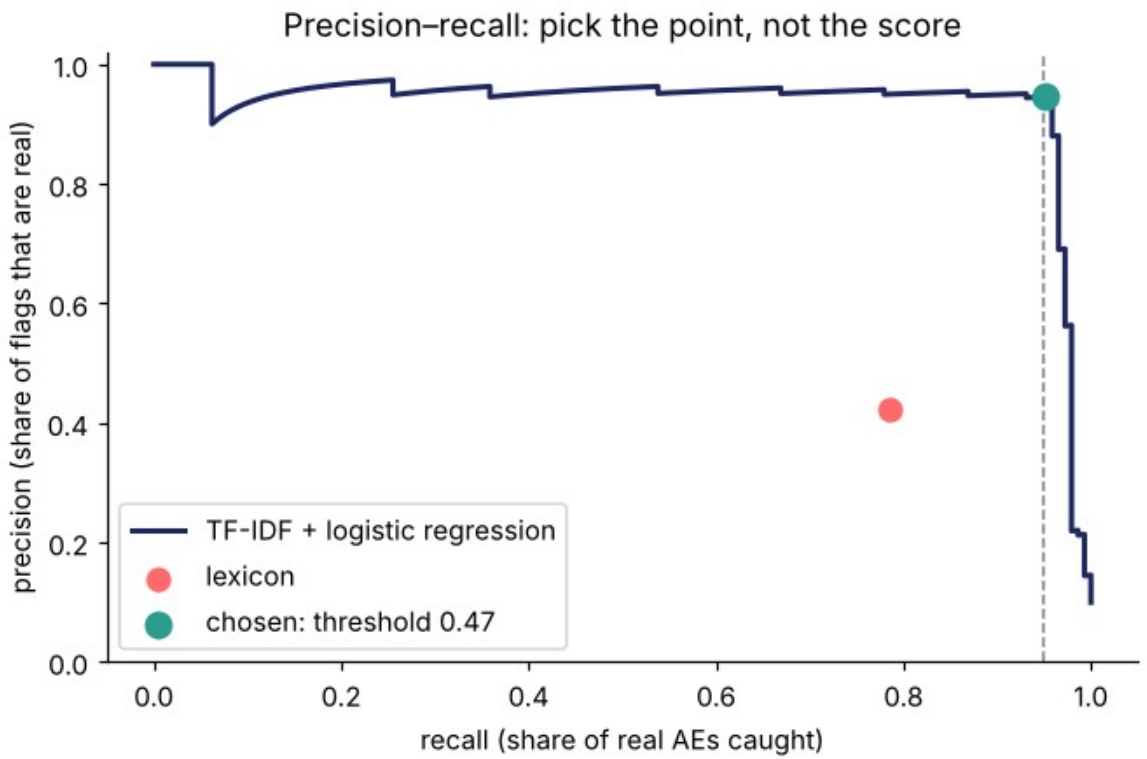
False negatives: event present, word missing

▸ “a 38yo male on metformin + SU w/ sugar lows, fyi”
▸ “burning when urinating after starting”
▸ “N/V”, “GI issues”, “nuasea”

The lexicon stays as the seriousness and expectedness lookup. It stops being the detector.

Source: Karimi et al. 2015. Numbers: notebook §3, reviewed notes.

TF-IDF + logistic regression: AUC 0.99, average precision 0.94



Recipe

▸ 1–2-grams, TF-IDF, L2 logistic regression, class weights. Negation scope: the three words after “no”, “denied” or “without” get a NOT_ prefix, so “no nausea” is a different feature from “nausea”.

	Precision	Recall	Flagged
Lexicon	42%	79%	270
LR at 0.50	94%	93%	—
LR at 0.47 (recall ≥ 95%)	95%	95%	146

Reviewed notes: 1,458, 5-fold CV. On the 2,742 unreviewed notes (instructor truth): recall 96%, precision 94%.

Source: Pang, Lee & Vaithyanathan 2002. Numbers: notebook §4.

3 · THRESHOLD

The threshold is a business decision

Threshold	Recall	Precision	Flags / week	Reviewer hours / week	AEs missed / year
0.20	97%	62%	13	1.3	12
0.30	97%	80%	10	1.0	14
0.47	95%	95%	8	0.8	20
0.50	93%	94%	8	0.8	29
0.70	79%	95%	7	0.7	89

- **Assumptions:** 6 minutes per note for a trained reviewer; about 84 notes a week; the reviewed sample represents the stream.
- **Choice:** at 95%+ recall the reviewer spends under an hour a week. Trading recall for that hour is not worth it.
- **The cost table goes into the model documentation;** it is what an inspector asks for.

Source: Numbers: notebook §4.

Five sentences every retrain has to pass

Probe	Tests	p(AE)
Patient developed nausea after starting GLUCOREN and stopped the drug.	plain positive	0.94
Patient had no nausea after starting GLUCOREN; tolerating well.	negation	0.34
HCP asked about the nausea rate in the label; no specific patient discussed.	general statement, no patient	0.07
One of his pts had N/V within 2 wks of starting, dose reduced.	abbreviations	0.99
A 61 y/o F was hospitalized with pancreatitis three weeks after starting GLUCOREN.	serious, unexpected event	0.87

Keep the list in version control and run it on every retrain.

Source: Ribeiro et al., ACL 2020 (CheckList). Numbers: notebook §5e.

04

Explaining the model

A safety reviewer acts on the span, not the score

4 · WHY

Why explanations are required here

"If you cannot explain how your AI came to a decision, it does not belong in a compliance system."

— IQVIA, Expert Predictions for 2026 (safety and pharmacovigilance)

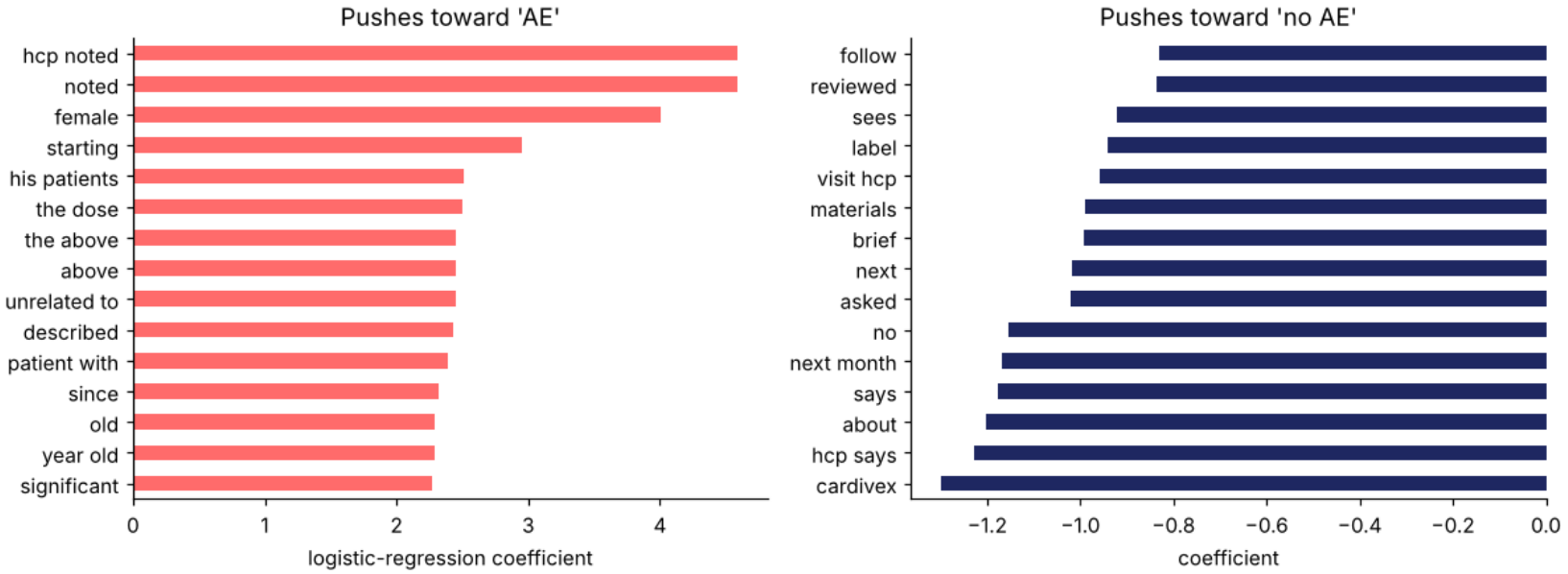
"The MAH will hold the ultimate responsibility for the AI that is used in their system and will need to document all AI usage information within the PV System Master File."

— IQVIA, Demanding More from AI Governance in Drug Safety

Question	Scope	Method	Reference
What does the model think an AE looks like?	global	coefficients	Molnar 2022
Why was this note flagged?	local, exact	coefficient × TF-IDF weight per word	—
Same, for a black box	local, model-agnostic	LIME	Ribeiro et al. 2016
Same, with consistent credit	local and global	SHAP	Lundberg & Lee 2017
Which tokens, for a transformer?	local	Integrated Gradients	Sundararajan et al. 2017; Jain & Wallace 2019

Source: IQVIA 2025 (both papers); Rudin 2019 on preferring interpretable models for high-stakes decisions.

What the model learned



Reading it

- Toward AE: event words and case words (“hcp noted”, “noted”, “female”, “starting”).
- Away: negation and generality (“cardivex”, “hcp says”, “about”, “says”).
- A payer, a region or a rep's habit in these panels is a shortcut to remove.

Source: Notebook §5a; pure function words hidden from the chart, kept in the model.

4 · LOCAL

Why this note was flagged

N01992 · p(AE) = 0.97 · flagged = 1 · lexicon = 1

Dr. **reports** Express Scripts is requiring prior authorization for GLUCOREN; two **pts** waiting over a week. Left PA form and hub phone number. **Also** metnioned **one of her pts dropped** off GLUCOREN **due** to **headache w/in 48 hours of starting.**

N03087 · p(AE) = 0.11 · flagged = 0 · lexicon = 1

HCP asked whether hypo is dose-related and whether **titration** helps. **Shared** the dosing **section of** the PI. Office staff friendly as always.

Coral pushes toward AE, navy away. Contribution = coefficient × TF-IDF weight, so the highlights are exact.

This is what goes in the reviewer's queue: the note with the span, not a probability.

Source: Notebook §5b.

LIME in four steps

1 · Perturb
switch words off at random; 400 neighbours

→

2 · Query
ask the model for p(AE) on each

→

3 · Weight
closer neighbours count more

→

4 · Fit
a weighted linear model; read its weights

"A 49 y/o m — constipated. Dose held, HCP to reassess." → *p(AE) = 0.95*

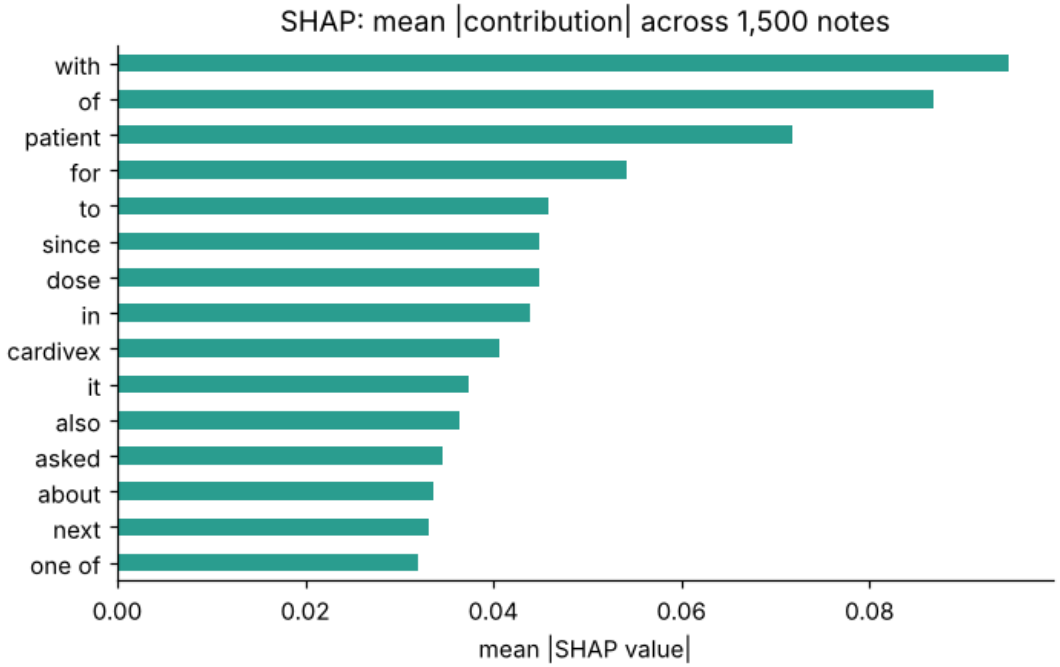
word	weight
held,	+0.132
reassess.	+0.097
Dose	+0.095
to	+0.063
HCP	+0.046
49	-0.056
y/o	-0.013

Cautions

- Local and unstable: change the seed and compare.
- Perturbed sentences are ones nobody writes; the model is being asked about inputs off its data.
- Works on any model, which is the point.

Source: Ribeiro, Singh & Guestrin, KDD 2016. Example: notebook §5c, fifteen lines of code.

SHAP, and a sanity check that found a shortcut



SHAP

- Shapley values: each feature's share of the prediction, with consistency guarantees. Mean |SHAP| ranks by how often a term matters as well as how strongly, so the ranking differs from the coefficients.

Payer names in the vocabulary

- |coef| > 0.5: “optumrx moved” (+2.0), “cigna is” (+1.8), “reports cigna” (+1.8), “optumrx” (+1.1).
- A shortcut from the training sample. Mask payer names → “payer”, refit: accuracy unchanged, vocabulary clean.

Explanations are for finding problems. Mask entities, refit, re-check.

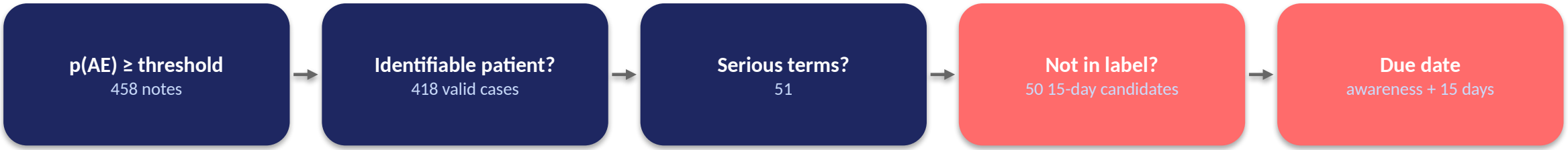
Source: Lundberg & Lee, NeurIPS 2017. Notebook §5d–5e. For transformers: Jain & Wallace 2019 (attention is not explanation); Sundararajan et al. 2017 (Integrated Gradients).

05

From flag to case

Rules after the model

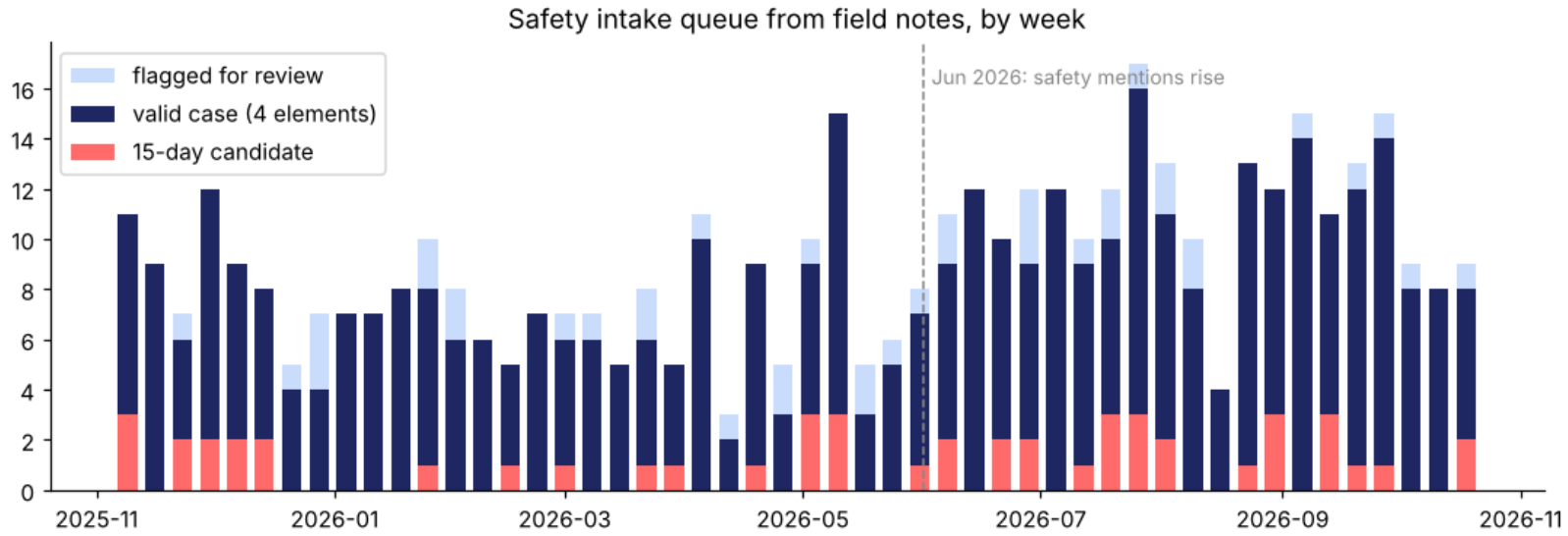
Rules on top of the classifier



Rule	How	Against instructor truth
Identifiable patient	regex: age/sex, “one of his patients”, a name, an MRN	precision 1.00, recall 0.93
Seriousness	lexicon: hospitalized, ER, ICU, died, pancreatitis…	—
Unexpected	lexicon column in_label = N	—
15-day candidate	valid and serious and unexpected	precision 1.00, recall 0.93 (50 of 51)
PHI in note	DOB, MRN, Mr./Mrs. + name, “patient First Last”	precision 1.00, recall 1.00

Source: Notebook §6. The missed 15-day cases were missed by the classifier upstream.

The intake queue, by week



Reading it

- About 9 flags a week, 0.9 reviewer-hours.
- 50 15-day candidates over the year.
- Safety mentions rise from June 2026.
- Reps are trained to report within one business day; the queue catches what the SOP missed, with the clock already running.

Source: Notebook §6.

Two rules the text brings with it

PHI in free text

- **Found:** 28 flagged notes with a name, DOB or MRN.
- **Rule:** HIPAA Safe Harbor lists 18 identifiers to remove before text is de-identified (45 CFR 164.514(b)).
- **Practice:** scrub at ingestion, and before any text leaves the firewall.

Off-label mentions

- **What:** unsolicited HCP questions about unapproved uses.
- **Rule:** document, route to Medical Information; the rep does not engage.
- **Model:** the same classifier family detects them (Field Guide exercise).

Source: 45 CFR §164.514(b).

06

Other uses of the same tools

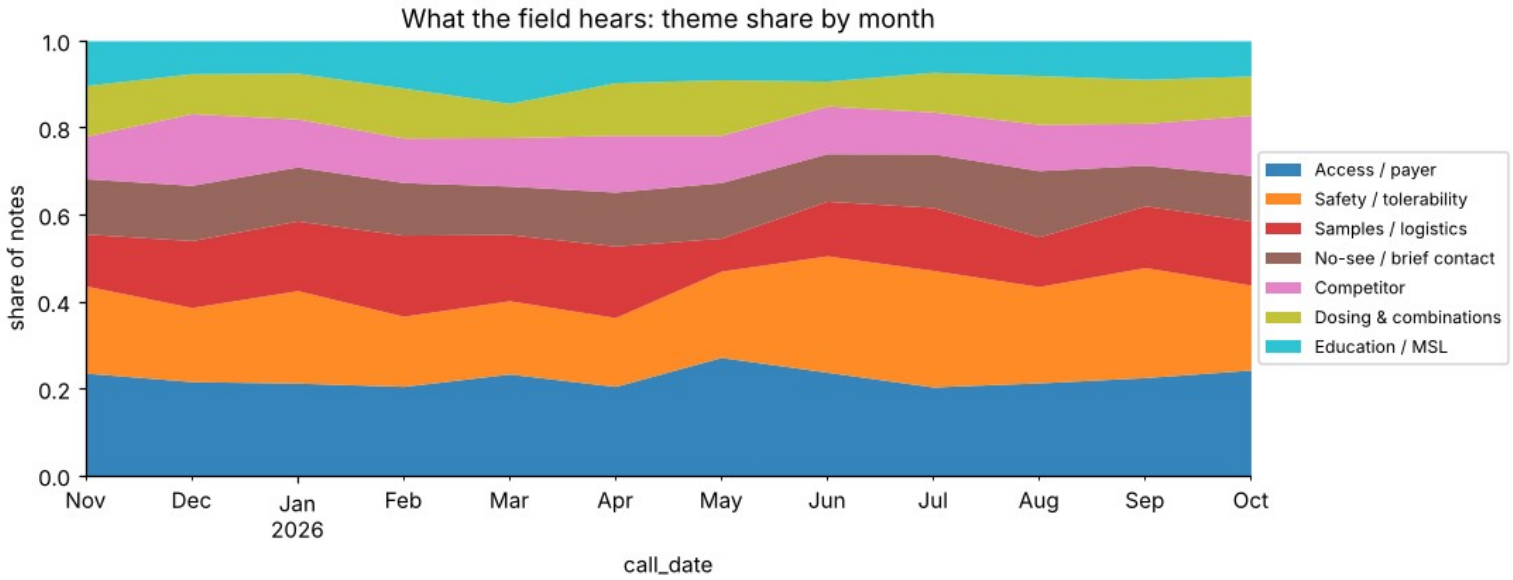
Topics, social listening, routing, signals

NLP use cases in pharma commercial

Use case	Text	Method	Output	Owner
AE detection	notes, posts, MI	classifier + rules	cases with due dates	Safety
What the field hears	call notes	topic model, rolled up	theme share by month, segment	Brand, insights
HCP sentiment	notes, surveys, ad boards	lexicon or classifier	negative share, drivers	Brand
Social listening	forums, social	classifier with own labels	tolerability themes, patient language	Insights, Safety
MI routing	inquiries	intent classifier	queue assignment	Medical Information
Semantic search	approved content, notes	embeddings	retrieval for reps, MSLs, MI	Medical, commercial ops
Literature and chart review	papers, EMR notes	screening models, NER	evidence tables, RWE variables	Medical Affairs
KOL profiling	publications, abstracts	NER, topics, networks	expert maps	Medical Affairs
Signal detection	the safety database	disproportionality	signals for committee review	PV

Source: IQVIA, AI in Medical Affairs (2025); IQVIA, Harnessing AI to Innovate Medical Information (2025).

Theme share by month



Reading it

- Safety share 18% before June, 25% after.
- Access is the largest theme (945 notes), 37% negative.
- NMF finds sub-themes (PA, copay, Medicare gap); we roll them up and assign safety with the detector.
- Competitor mentions track the CARDIVEX label update (notebook §7).

Source: Lee & Seung 1999 (NMF); Blei, Ng & Jordan 2003 (LDA); Grootendorst 2022 (BERTopic); Hutto & Gilbert 2014 (sentiment lexicon). Notebook §7.

The same detector on patient posts

Model	Precision	Recall
Trained on rep notes, scored on posts	64%	91%
Lexicon on posts	64%	85%
Trained on post labels (5-fold CV)	94%	99%

Source	Posts	Flagged	Serious terms	Duty
Aspen community (sponsored)	1,201	311	43	screen regularly
T2D Circle forum	1,799	452	90	act on awareness

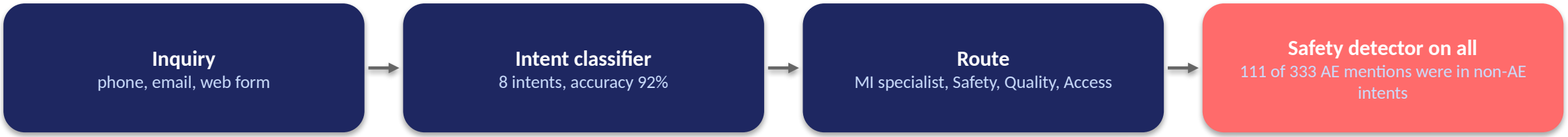
From the literature

- Social-media AEs skew to mild, symptom-level events and under-represent serious ones (Golder et al. 2015).
- Twitter event rankings correlated with FAERS; counts did not (Freifeld et al. 2014).
- Lay language is the main difficulty; embeddings help (Nikfarjam et al. 2015).

Label the new stream before trusting a model trained on another one.

Source: Golder, Norman & Loke 2015; Freifeld et al. 2014; Nikfarjam et al. 2015; Sarker et al. 2015.

Medical-information routing



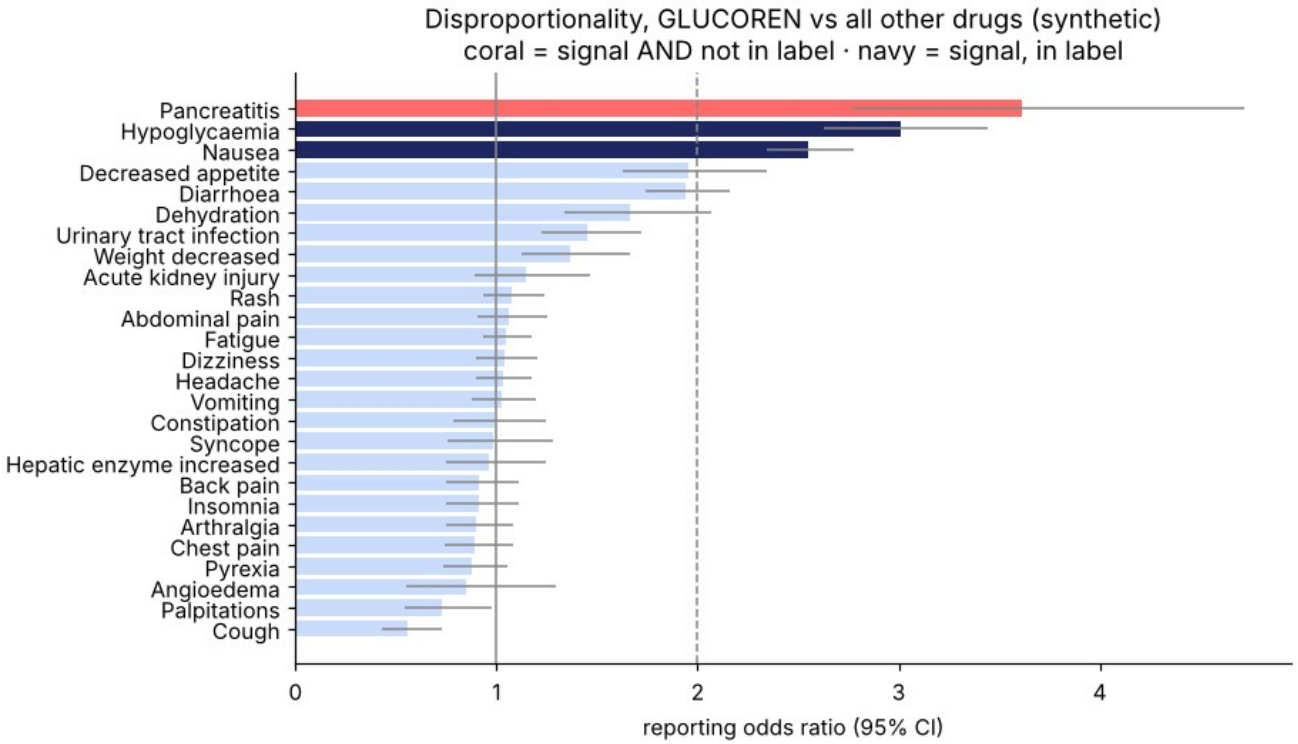
> *"…the automatic transfer of adverse events (AEs) and product complaints (PQCs) to their requisite systems, which helps maintain compliance amid spikes."*
>
> — IQVIA, Harnessing AI to Innovate Medical Information (2025)

> *"In medical information, most data is spontaneous and unstructured. That implies a huge responsibility for our confidentiality and how we handle that data."*
>
> — IQVIA, Expert Predictions for 2026

A dosing call that mentions nausea is a dosing call and a case.

Source: Notebook §9.

From cases to a signal



PRR and ROR

- PRR = (a/(a+b)) ÷ (c/(c+d)); ROR = ad/bc with a 95% CI. Evans's screen: PRR ≥ 2, χ^2 ≥ 4, n ≥ 3. Bayesian shrinkage (BCPNN) for sparse cells.

Synthetic FAERS-style counts

- Signals: Pancreatitis, Hypoglycaemia, Nausea.
- Pancreatitis, PRR 3.6, not in the label: signal-review committee.
- No exposure denominator and heavy under-reporting: a hypothesis, not a rate.

Source: Evans, Waller & Davis 2001; Bate et al. 1998; Harpaz et al. 2012. Notebook §10; search with LSA/embeddings in §11.

07

Close

GenAI in one slide, governance, exercise

Large language models change the cost of this pipeline, not its logic

What changes

- **Zero-shot extraction:** a prompt returns the four elements from a note with no training set.
- **Normalization and narrative:** lay term to coding dictionary; case narratives; weekly summaries of field notes.
- **Retrieval plus generation:** answers grounded in approved content.

What does not

- **The target:** four elements, seriousness, the 15-day clock.
- **The human:** every case is reviewed.
- **The data rules:** PHI scrubbed before text leaves the firewall.
- **The evidence:** precision, recall and workload on your own labelled data.

>70%
of GenAI experiments never reach production

The GenAI session covers prompting and evaluation, hallucination, RAG over MLR-approved content, cost, and the FDA's 2025 draft guidance on AI credibility. Tonight's detector is the baseline.

Source: IQVIA, Where to Deploy Generative AI Solutions in Life Sciences (2025); FDA draft guidance, January 2025.

7 · GOVERNANCE

What the PV lead will ask for

Item	Write down
Context of use	which stream, which decision, who acts, what happens on error
Data	source, period, reviewer labels and their noise, PHI handling
Performance	AUC and AP for comparison; the operating point with recall, precision, workload; held-out and behavioural tests
Explanations	global review, per-flag spans, leak checks
Human oversight	who reviews every flag; escalation; the SOP it sits beside
Monitoring	flag-rate drift, probes on every retrain, domain shift on new streams
Record	the PV System Master File

Source: FDA / Health Canada / MHRA, Good Machine Learning Practice (2021); FDA draft guidance (Jan 2025); IQVIA (Raizada) 2025.

7 · EXERCISE

In-class exercise, 25 minutes, pairs

A · Eight notes by hand (10 min)

▸ AE? four elements? serious? unexpected? due date? Write the phrase that decided it.

B · Read an explanation (8 min)

▸ Two word-contribution outputs. Trust the flag? Send the model back? Which probe to add?

C · Set the threshold (7 min)

▸ Reviewer budget of 2 hours a week: pick the threshold, state the misses, justify it to Safety. Then triple the volume.

When	What
Mon Nov 2	Academic holiday. Homework 2 due 11:59 PM ET
Mon Nov 9	Midterm, closed book, Weeks 1–7. Study guide posted a week ahead
Mon Nov 16	Guest speaker: agentic AI in pharma commercial. Two questions in advance